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Mode II and Mixed Mode Delamination Growth in Composite Materials Using Peridynamic Theory

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This paper presents a method using Peridynamic Theory (PD) to model delamination failure in a unidirectional (UD) composite specimen under Mode II and mixed mode loading conditions. The original brittle damage model in PD is modified as a bilinear law in order to capture progressive softening. The proposed constitutive relations are implemented in bond-based PD formulation under plane stress conditions. The PD model of the End Notched Flexure (ENF) and Mixed Mode Bending (MMB) test specimens are generated using a MATLAB code. The PD model of the specimens are implemented in Finite Element Analysis (FEA) software ABAQUS. PD solutions of force displacement curves are obtained with explicit time integration scheme in ABAQUS. The PD solutions are verified using the analytical and FEA results from the literature and the analytical solutions of ENF and MMB problems using Corrected Beam Theory (CBT). In addition to PD solutions, FEA of the ENF and MMB problems are performed using Cohesive Zone Modelling (CZM) approach. It is observed that the PD results obtained with bilinear material law are in good agreement with both FEA, analytical results and the results from the literature.

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Keywords: Mode II loading; Mixed Mode loading; Cohesive Zone Method; Peridynamic Theory; ENF Test; MMB Test**Nomenclature**

CBT	Corrected Beam Theory	FEA	Finite Element Analysis
CCM	Classical Continuum Mechanics	MMB	Mixed Mode Bending

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CZM	Cohesive Zone Method	PD	Peridynamics
DCB	Double Cantilevered Beam	SBT	Simple Beam Theory
ENF	End Notched Flexure	UD	Unidirectional

1. Introduction

In recent years, there has been an increase in the usage of composites materials. These materials are mostly used in aerospace, defense, and automotive industries to take advantage of their high strength, stiffness, and low weight. Internal failure of composites occurs due to breaking of fibers, development of micro cracks in matrix, debonding between fibers and matrix and delamination. Delamination in composites and fracture of adhesive joints are the most common failure types. The structural strength and stiffness decrease with the result of these defects. The failure occurs because of the separation at the interface region Fan et al. (2008).

Interface strength of the material is essential for failure analysis and the characterization of delamination failure in composites. Various numerical methods have been developed to investigate failure behavior of engineering materials such as cohesive zone method (CZM) Hillerborg et al. (1976), Xie and Waas (2006) and extended finite element method Moës and Belytschko (2002), Fries and Belytschko (2012).

CZM provides solution for problems related to crack initiation and propagation. It is firstly introduced by Dugdale (1960) with a thin plastic zone that is generated in front of the notch. Following his work, Barenblatt (1962) introduced cohesive forces on a molecular scale in the zone that Dugdale had pointed out in order to solve the problem of equilibrium in elastic bodies with cracks. Later, many researchers worked on CZM approach, Hillerborg et al. (1976), Turon et al. (2007). In this approach, material behavior within the damage zone explained with a traction-separation law that is also known as cohesive law. In this law, a damage zone is developed in the cohesive layer. The damage starts to develop when the stress limit is reached, and the stress decreases as the damage grows. Finally, the stress becomes zero when the separation reaches a critical value. The relation between the stress and displacement is governed by the cohesive law and the area under stress-displacement curve gives the critical strain energy release rate Blackman et al. (2003), Fan et al. (2008). This law is appropriate for different types of loading conditions such as: Mode I, Mode II, and mixed mode loadings.

Silling proposed a new theory called Peridynamic theory (PD) in 2000 which is based on modelling the material points and their interactions within a finite region Silling (2000). This theory is based on integral equations rather than differential equations in Classical Continuum Mechanics (CCM). These equations are valid at the presence of discontinuities in the structures such as voids and cracks. As opposed to CCM, crack propagation can be predicted without any external failure criteria. At present, there are many different PD formulations and bond-based PD modelling is the original and most commonly used PD formulation Silling et al. (2007). This method can be applicable for various materials including concrete Tong et al. (2020), glass Kilic and Madenci (2009), composites Oterkus and Madenci (2012) but poisson's ratio must be fixed at $\nu = 0$ for plane stress problems and at $\nu = -1$ for plane strain problems Huang et al. (2015). Using bond-based PD formulations, micro elastic damage model for prototype micro elastic brittle materials introduced by Silling and Askari (2005). In their study, bond force is given as a linear function of bond stretch until a critical stretch value where the bond force sharply decreases to zero. Therefore, softening behavior was not considered. A new PD bond model for Mode I and mixed mode loading is applied by Tong et al. (2020) for concrete. In the developed damage model, bond force increases linearly with the bond stretch up to critical stretch value. Then, exponential softening behavior is applied considering residual strength of the bonds.

In this paper, a new bond-based model is proposed to model the failure behavior in composite UD specimen based on bilinear failure model. ENF and MMB unidirectional composite test specimen data obtained from Turon et al. (2010) is used for PD modelling and CZM approach. PD modelling of test specimens are generated using MATLAB pre-processor code. Obtained PD results have a good agreement with the results obtained from analytical solutions, FEA and the numerical results of Turon et al. (2010).

2. Numerical Methods

Many important concepts are developed to predict crack initiation and propagation in composite materials. In the literature, CZM is developed to predict failure behavior of materials using cohesive zone elements that are placed along the element boundaries. Its formulations can be easily implemented to finite element codes. Therefore, crack growth occurs only between regular elements. However, crack paths are highly sensitive to mesh texture. In order to overcome these difficulties, a PD model that is one of the non-local theories is developed. In this part of the study, CZM and PD methods will be explained in detail.

2.1. Cohesive Zone Method

The CZM is a common method in the literature to model initiation and propagation of delamination Camanho et al. (2003), Alfano (2006). In CZM, traction separation laws are employed where tractions of the crack tip are related to the crack tip displacements. For the crack growth analysis in CZM, constitutive laws with trapezoidal de Moura et al. (2008), cubic polynomial Blackman et al. (2003), exponential van den Bosch et al. (2006) or linear/polynomial Pinho et al. (2006) are commonly used. It is observed that bilinear law is adequate to capture delamination behavior in composite materials Heidari-Rarani and Ghasemi (2017). Therefore, a bilinear cohesive law shown in Fig. 1. is applied for modelling the delamination in the current study.

Bilinear relation can be investigated in three parts. The first part is called as the intact region which indicates the undamaged part and goes up to initiation criterion point (Δ^0, σ^0) . The slope of the intact part is known as penalty stiffness K and its value must be high enough to prevent interpenetration of the crack faces from the cohesive elements Song et al. (2008). In this study, it is taken as 10^6 N/mm. The second part is called as the softening region corresponding to region between delamination initiation Δ^0 and delamination propagation Δ^f . After Δ^f , the interface fails completely, and cohesive elements are traction free.

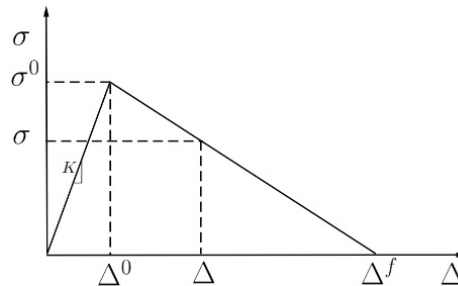


Fig. 1. Traction Separation law.

2.2. Peridynamic Method

PD theory is a non-local continuum theory with the advantage of modelling discontinuities and cracks Silling (2000), Oterkus and Madenci (2012). In PD, the material point $\mathbf{x}$ interacts with other material points within a distance δ , as shown in Fig. 2. Equation of motion in PD formulation can be written as:

$$\rho(\mathbf{x})\ddot{\mathbf{u}}(\mathbf{x},t) = \int_H \mathbf{f}(\mathbf{u}' - \mathbf{u}, \mathbf{x}' - \mathbf{x})dH + \mathbf{b}(\mathbf{x},t), \quad (1)$$

where ρ is the mass density, $\mathbf{u}$ is the displacement vector field, $\mathbf{f}$ is the pairwise force density function and $\mathbf{b}$ is the body force density. In PD theory, $\mathbf{f}$ is a function of the relative position vector, $\boldsymbol{\xi} = \mathbf{x}' - \mathbf{x}$ and the relative displacement vector, $\boldsymbol{\eta} = \mathbf{u}(\mathbf{x}',t) - \mathbf{u}(\mathbf{x},t)$ between the two material points and it is derived from micro elastic potential, w as Silling and Askari (2005):

$$\mathbf{f}(\boldsymbol{\eta}, \boldsymbol{\xi}) = \frac{\partial w(\boldsymbol{\eta}, \boldsymbol{\xi})}{\partial \boldsymbol{\eta}}. \quad (2)$$

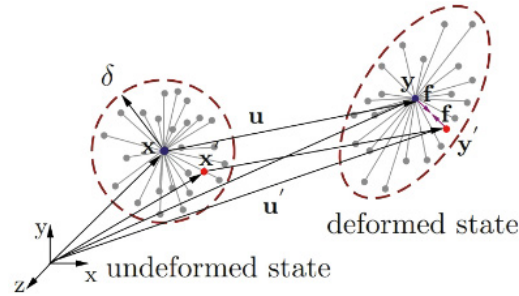


Fig. 2. Representation of bond-based PD model.

The micro potential energy is the energy in a single bond at a given point. It depends on the relative displacement vector $\boldsymbol{\eta}$ through the distance between the deformed points. This scalar valued function can be expressed as:

$$w(\boldsymbol{\eta} + \boldsymbol{\xi}, \boldsymbol{\xi}) = w(\boldsymbol{\eta}, \boldsymbol{\xi}). \quad (3)$$

Combining Eq. (2) and (3) and differentiating with respect to $\boldsymbol{\eta}$ shows that the PD force density vector, $\mathbf{f}$ is a function dependant on the reference position vector $\boldsymbol{\xi}$, relative displacement vector $\boldsymbol{\eta}$, bond constant of the material c , and the value of stretch s in the absence of thermal loading. This relation is given in Eq. (4).

$$\mathbf{f}(\boldsymbol{\eta}, \boldsymbol{\xi}) = \frac{\boldsymbol{\eta} + \boldsymbol{\xi}}{|\boldsymbol{\eta} + \boldsymbol{\xi}|} c s, \quad (4)$$

where c is called as the PD bond constant of the material and s is the stretch of the bond and may be calculated as:

$$s = \frac{|\boldsymbol{\eta} + \boldsymbol{\xi}| - |\boldsymbol{\xi}|}{|\boldsymbol{\xi}|}. \quad (5)$$

The PD bond constant c can be obtained by equating the strain energy density for PD theory and the strain energy density for CCM. As a result of this, c may be obtained as:

$$c = \frac{18K}{\pi\delta^4}, \quad (6)$$

where K is the bulk modulus of the material, and δ is the radius of the horizon. PD formulation is implemented in ABAQUS using truss elements (T3D2) that represent the PD bonds as suggested by Macek and Silling, (2007). Mesh of truss elements are generated with an in-house code developed in MATLAB. Elastic modulus (E_t) and cross-sectional area of truss elements (A_t) are taken as in Macek and Silling (2007):

$$E_t = c\Delta x^4, \quad A_t = \Delta x^2, \quad (7)$$

where Δx indicates the grid size.

2.2.1. PD Interface Failure Model

Material failure can be considered in PD theory by breaking the bonds when the stretch between material points reaches a critical value. The material model in the original PD formulation Silling and Askari (2005) is brittle (see the black line in Fig. 3a). However, the interface failure problem requires a softening behaviour that is considered in traction separation laws in CZM. Therefore, the original PD material model is modified as a bilinear model (see the red line in Fig. 3a).

In the modified formulation, the micro potential at the failure can be calculated as:

$$w = w_0 + w_1, \quad (8)$$

where w_0 and w_1 are given in Eq. (9) and Eq. (10):

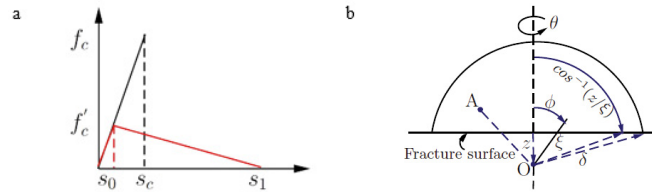


Fig. 3. Peridynamic failure model a) bond force versus stretch b) calculation of energy break bonds per unit area

$$w_0 = \int_0^{s_0} f ds = \frac{cs_0^2 \xi}{2}, \quad (9)$$

$$w_1 = \int_{s_0}^{s_1} f ds = \frac{cs_0^2 \xi}{2} (s_1 - s_0). \quad (10)$$

Substituting Eq.(9) and Eq.(10) into Eq. (8) w can be calculated as:

$$w = \frac{cs_0 s_1 \xi}{2}, \quad (11)$$

where the parameters s_0 and s_1 denotes the critical stretch values at the beginning of softening stage and final failure, respectively. It is possible to link the energy required to break all the bonds per unit area (as given in Fig. 3b) to the macroscopic fracture toughness G_C . Therefore, the following relation between the G_C and w in can be written as:

$$G_C = \int_0^\delta \int_0^{2\pi} \int_0^{\cos^{-1}(z/\xi)} \int_0^{\cos^{-1}(z/\xi)} w \xi^2 \sin \phi d\phi d\xi d\theta dz. \quad (12)$$

By evaluating the integral in Eq. (12), the fracture toughness can be obtained in terms of s_0 and s_1 as:

$$G_C = \frac{cs_0 s_1 \pi \delta^5}{10}. \quad (13)$$

In order to determine s_0 and s_1 , it is assumed that the ratio of Δ^f / Δ^0 in Turon et al. (2010) is the same with s_1 / s_0 in the modified PD material model.

3. Problem Description

In the current paper, static solutions of ENF and MMB test specimens are considered by using analytical, PD and FEA method. The ENF and MMB tests are explained and the analytical solutions are found using Corrected Beam Theory (CBT) given in the following subsections.

3.1. The ENF Test

Mode II interlaminar fracture toughness G_{II} can be determined conducting an ENF test ASTM D7905 which is carried out with a simple three-point bending specimen that has an interface crack length a . In ENF test, the specimen is constraint with a pinned support at one end and a roller support at the other end. The load P is applied at the mid-span as shown in Fig. 4.

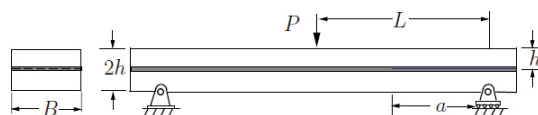


Fig. 4. Representation of ENF test.

The relation between G_{II} and P can be written using Linear Elastic Fracture Mechanics as follows:

$$G_{II} = \left(\frac{P^2}{2B} \right) \frac{dC}{da}, \quad (14)$$

where B is the width, and $C = \delta / P$ is the compliance of the specimen. Analytical solution of ENF problem can be obtained with several theories such as: Corrected Beam Theory (CBT), Simple Beam Theory (SBT) and Compliance Calibration Method. Moura and Morais (2008) stated that CBT gave better results since the mentioned method considers shear and beam rotation at the crack tip. For CBT, expression for compliance C is obtained using Timoshenko beam theory Wang and Williams (1992) and can be calculated as:

$$C = \frac{3(a + \Delta_{II})^3 + 2L^3}{8BE_{11}h^3} + \frac{3L}{10G_{13}Bh}, \quad (15)$$

where E_{11} is the longitudinal modulus of elasticity, G_{13} is the out of plane shear modulus, L is the half length of the specimen and h is the half thickness of the specimen. Δ_{II} in Eq. (15) denotes the correction factor for Mode II loading due to the beam rotation. In the literature, it is calculated as $\Delta_{II} = 0.42\Delta_I$ Wang and Williams (1992) where Δ_I is the correction for Mode I and can be calculated as Fernández et al. (2013):

$$\Delta_I = \left(\frac{E_{11}}{11G_{12}} \right)^{\frac{1}{2}} \left[3 - 2 \left(\frac{1.18\sqrt{E_{11}E_{22}}}{G_{12} + 1.18\sqrt{E_{11}E_{22}}} \right)^2 \right]^{\frac{1}{2}} h, \quad (16)$$

where G_{12} is the in plane shear modulus, and E_{22} is the transverse modulus of elasticity. Substituting Eq. (15) into Eq. (14), fracture toughness for Mode II loading G_{II} might be obtained as:

$$G_{II} = \frac{9P^2(a + 0.42\Delta_I)^2}{16B^2E_{11}h^3} F_v, \quad (17)$$

where F_v represents the correction factor for the large displacement of the end blocks. The value of F_v might be calculated using the formulations using large beam deflection theory given in Feito (2012).

3.2. The MMB Test

The MMB test is used to determine interlaminar fracture toughness G of continuous fiber reinforced composite materials at various Mode I and Mode II loading ratios ASTM D6671M. The MMB test specimen is a combination of Double Cantilevered Beam (DCB) and ENF specimens as shown in Fig. 5.

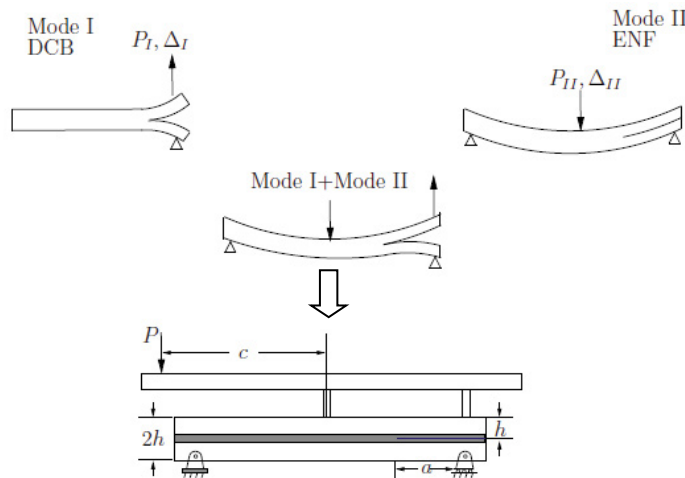


Fig. 5. Representation of MMB test.

In an MMB test, a lever is used to load the specimen. Applying a load at the end of the lever, P_I and another load, P_{II} is applied at the center of the specimen a mixed mode loading condition is obtained. Note that upward force P_I creates Mode I loading, while the downward force P_{II} creates Mode II loading for the laminate. Similar to ENF test, the specimen has a pinned support at one end and a roller support at the other end in an MMB test.

Mixed mode ratio, B can be controlled by the length of the lever arm position c as shown in Fig. 5 and can be calculated as:

$$B = \frac{G_{II}}{G} = \frac{G_{II}}{G_I + G_{II}}, \quad (18)$$

where, G_I is the Mode I fracture toughness and G is the total fracture toughness. Using this ratio, Benzeggagh and Kenane (1996) suggested a failure criterion to calculate critical fracture toughness by curve fitting for G_c vs B curve as follows:

$$G_c = G_{IC} + (G_{IIC} - G_{IC})B^\eta, \quad (19)$$

where, G_{IC} is the Mode I critical fracture toughness, G_{IIC} is the Mode II critical fracture toughness and η is the factor that is found by curve fitting. Analytical solution of MMB test can be obtained using CBT. Kinloch et al. (1993) proposed that CBT can be developed considering the increasing length of delamination. For Mode I and Mode II loading conditions, the fracture toughness values can be calculated as Reeder (2003):

$$G_I = \frac{12P_I^2}{B^2E_{11}h^3}(a + \chi h)^2, \quad G_{II} = \frac{9P_{II}^2}{16B^2E_{11}h^3}(a + 0.42\chi h)^2, \quad (20)$$

where, P_I and P_{II} are the Mode I and Mode II partitions of the main load and χ is the crack length correction parameter. The detailed calculations for P_I and P_{II} are given in Reeder (2003). In Eq. (21) χ can be calculated as:

$$\chi = \sqrt{\frac{E_{11}}{11G_{13}}} \left\{ 3 - 2 \left(\frac{\Gamma}{1 + \Gamma} \right)^2 \right\}, \quad (21)$$

where Γ is the transverse modulus correction parameter and can be obtained using Eq. (22):

$$\Gamma = 1.18 \frac{\sqrt{E_{11}E_{22}}}{G_{13}}. \quad (22)$$

Finally, combining individual modes given in Eq. (20), the total fracture toughness, G can be obtained as:

$$G = G_I + G_{II}. \quad (23)$$

4. Results

In this study, PD simulations of ENF and MMB specimens in Turon et al. (2010) are performed with a bilinear PD material law. The same specimen is used for both ENF and MMB tests. The geometrical properties of the specimen shown in Fig. 4 are given in Table 1 and its material properties are given in Table 2. Note that the specimen is made of carbon-fiber reinforced epoxy composite.

Table 1. Geometrical properties of ENF and MMB specimens

L	B	h	a
75 mm	20 mm	1.55 mm	35 mm

Table 2. Material properties of carbon-fiber reinforced epoxy composite (Turon et al., 2010)

E_{11}	$E_{22} = E_{33}$	$G_{12} = G_{13}$	G_{23}	$\nu_{12} = \nu_{13}$	ν_{23}
120.0 GPa	10.5 GPa	5.25 GPa	3.48 GPa	0.3	0.5
$\sigma_n^{\max}$	$\sigma_s^{\max}$	G_{IC}	G_{IIC}	K	η
30 MPa	58.9 MPa	0.260 N/mm	1.002 N/mm	10^6 N/mm	2.0

The PD models of the specimens are generated using truss elements with an in-house script developed in MATLAB as shown in Fig. 6. Bilinear material law is used for the interface bonds that are shown with the red color in Fig. 6. The FEA models are created using two dimensional (CPE4R) elements and connected with two-dimensional cohesive layer COH2D4 elements in ABAQUS (2019). Mesh size is taken as 0.32 mm in FEA calculations.

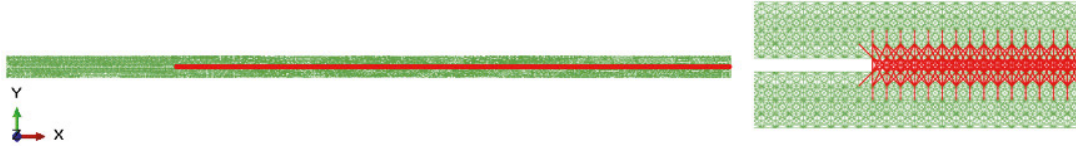


Fig. 6. PD bonds for the lamina (green) and the interface bonds (red).

The PD solutions of the force displacement curves for the ENF and MMB specimens are compared with the results obtained using with CZM in ABAQUS, and numerical results in Turon et al. (2010) and the analytical solutions explained in Sections 3.1. and 3. 2.

4.1. ENF Problem

The PD solution of ENF test is performed by applying displacement at the mid-span of the specimen and boundary conditions applied as given in Fig. 4. The locality of the PD interactions is affected by the horizon size $\delta = m\Delta x$ where m is horizon size constant. Therefore, the effect of horizon size, $\delta = m\Delta x$ on the force displacement result of the ENF test is investigated first. In the m convergence study, three different horizon sizes with $m = 3.015, 5$, and 10 are used and grid size is fixed at $\Delta x = h/5$. Force displacement curves of m convergence study is depicted in Fig. 8a. It is observed that $m = 3.015$ gives nearly identical results with the numerical data in Turon et al. (2010). As the m increases, the discrepancy between the numerical results of Turon et al., (2010) and PD solution increases due to the excessive wave dispersion which is discussed in Silling and Askari (2005). For further investigations the m value is fixed at 3.015 for based on the m convergence study. Next, the effect of mesh size, Δx on the PD results are investigated. PD models with mesh densities $\Delta x = h/3, h/5$ and $h/7$ as shown in Fig. 7 are generated for δ convergence study.

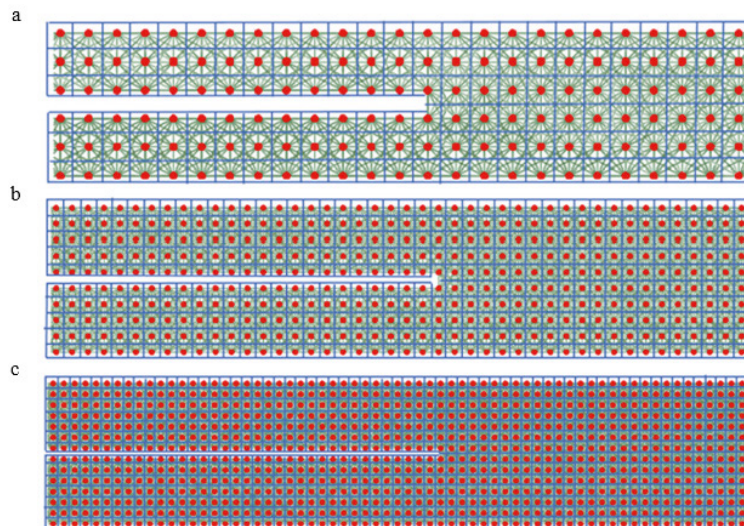


Fig. 7. PD models created for different grid sizes, a) $\Delta x = h/3$ b) $\Delta x = h/5$ c) $\Delta x = h/7$.

The force displacement results obtained with three different mesh sizes are shown in Fig. 8b. The obtained results for $\Delta x = h/3, h/5$ and $h/7$ are nearly identical with the numerical results in Turon et al. (2010). It can be noted that, the mesh convergence is obtained for the mesh sizes smaller than $\Delta x = h/5$. PD mesh size is taken as $\Delta x = h/5$ for the further investments for computational efficiency.

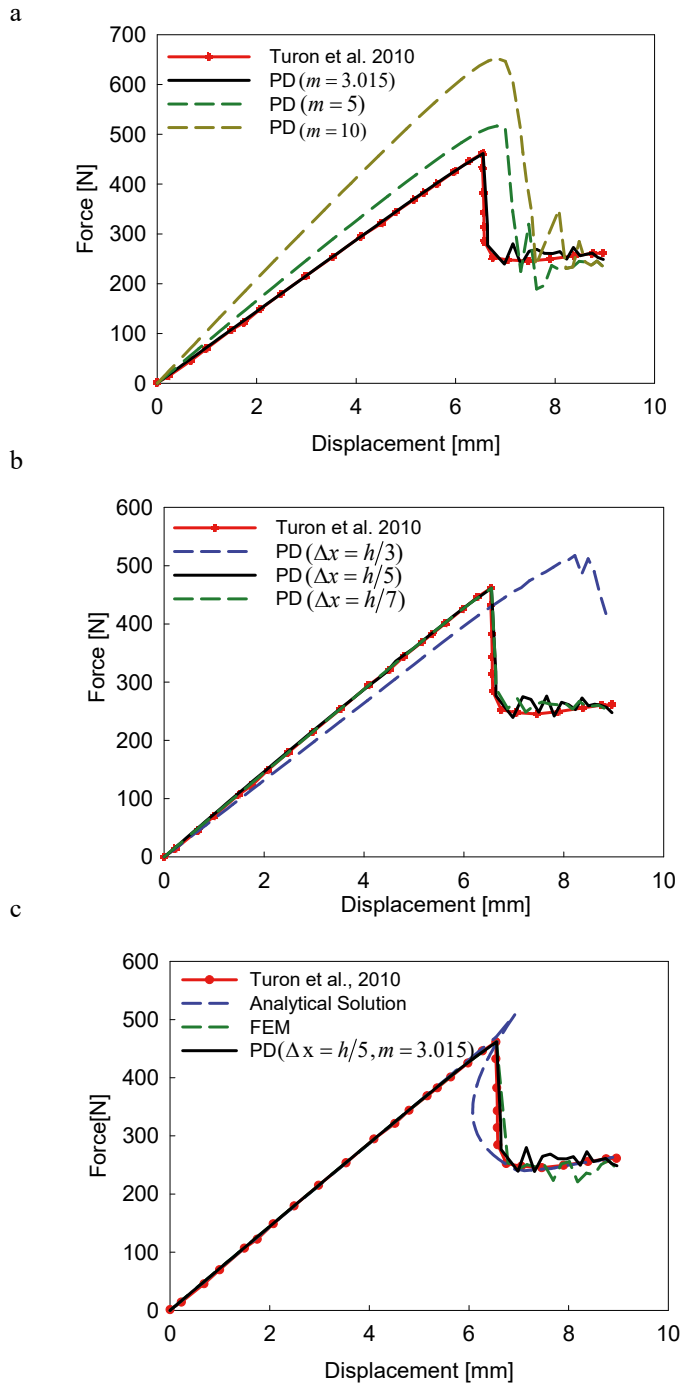


Fig. 8. Force-displacement results of a) m convergence study b) δ convergence study c) comparison of PD, CZM and analytical solutions.

PD results of ENF problem using optimum horizon and mesh sizes are compared with FEA results, analytical solution with CBT and numerical results of Turon et al. (2010) in Fig. 8c. PD results are in very good agreement with numerical results of Turon et al. (2010), FEA and analytical solutions in linear region. Although there is a small deviation with the analytical solution after the failure initiation, the results of PD and FEA are nearly identical with the numerical solution results of Turon et al., (2010) after failure initiation as seen in Fig. 8c.

4.2. The MMB Problem

In the MMB test, the same specimen with the ENF test is considered. The positions of the Mode I and Mode II partitions of the load P in Fig. 5 are adjusted according the mode ratios of 20%, 50%, 70% and 80% . Mode ratio of 20% is applied by setting the arm distance (c) to 149.05 mm. Mixed mode ratios of 50%, 70% and 80% results are obtained by setting the (c) to 63.18 mm, 46.71 mm, 41.06 mm respectively. These (c) values are determined using the equations given in ASTM D6671M (2006). The boundary conditions are applied as shown in Fig. 5. For PD modelling of the MMB specimen the model is created with $m = 3.015$ and $\Delta x = h/5$. Force displacement curves obtained from PD method are compared with the results of FEA and the analytical solution. These results are depicted in Fig. 9.

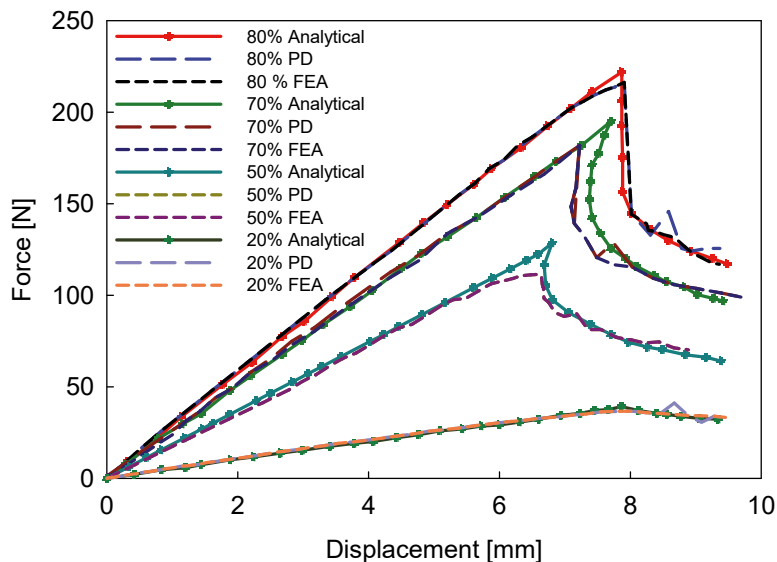


Fig. 9. Comparison of force-displacement results of MMB problem.

These results show that in the linear region PD formulation results have good agreement with analytical and FEA results although there are minor differences after the failure initiation.

5. Conclusion

In this study, analytical solution, and numerical simulations of ENF and MMB test specimens were performed. For FEA of the specimen CZM is used and two-dimensional cohesive elements are created in ABAQUS. Then, bond-based PD models are implemented using a new modified bilinear material law. Obtained PD results indicates that this new PD model with bilinear constitutive relation can correctly capture the numerical results of Turon et al. (2010). Note that PD results are also validated using CZM and analytical solutions.

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